ACCESS · INDIGENOUS SKILLS 4 SUCCESS

Worked answers and teaching guidance

Volume 1 · Individual evidence, checks and next practice

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Contents

1. Unit 1 · Answer and teaching guidance
2. Unit 2 · Answer and teaching guidance
3. Unit 3 · Answer and teaching guidance
4. Unit 4 · Answer and teaching guidance
5. Unit 5 · Answer and teaching guidance
6. Unit 6 · Answer and teaching guidance
7. Additional worked answers and assessment guidance
8. Coverage and delivery record

Unit 1 · Answer and teaching guidance

Use 10 minutes for source/measure vocabulary, 20 for “Report what the data can support” through “Retrieve and preserve a source record”, 25 for “Match a question to a document” through “Check the funding statement”, 25 for “Make the strongest supported comparison” through “Revise a claim when the scope changes”, and 10 for revision. Prepare I1–I5 on paper and an actual device turn for each learner. The public-source register and retained extracts make the lesson workable offline; online retrieval remains unobserved until performed.

Report what the data can support

Expected evidence and useful feedback: Employment is 246,750 people in B.C. construction in 2024. Full-time share is 12 percentage points above the all-industry figure. 87,280 is forecast construction openings over 2025–2035, not guaranteed paving jobs. I1b's categories sum to 427 statistical locations; `175 − 63 = 112 locations`; `175 ÷ 427 × 100 = 40.983606…%`, or **40.98%**. The road-construction classification includes highways, streets and bridges. Locations are not unique companies or workers; size ranges must not be used to estimate total employees. Comparing this location count with all-construction employment mixes measure, scope and period.

Retrieve and preserve a source record

Expected evidence and useful feedback: A retained note identifies title, WorkBC/Province of B.C., URL, access date, metric/year and any change. Observe locating, saving and reopening independently of the accuracy of the paper answer.

Match a question to a document

Expected evidence and useful feedback: I2 supports selected orientation requirements. I3 explains the document system, but the actual signed contract and relevant amendments/special provisions are needed to establish its applicable edition. I4 identifies the public funders and contractor. I3b confirms End Product Specification in the 2006 clause; it does not establish every old note’s intended usage. Profit and a separate unexplained note still require further evidence. A register must not silently label a specification “law.”

Check the funding statement

Expected evidence and useful feedback: `215.19 + 385.58 = $600.77 million`. Federal fraction `215.19 ÷ 600.77 × 100 = 35.819032…%`, or **35.82%**. The provincial share is **64.18%** after rounding. The total exceeds $600 million by $0.77 million. Province is the recorded contracting authority; Canada and B.C. are reported funders; Kicking Horse Canyon Constructors is the awarded contractor. The source does not identify profit.

Make the strongest supported comparison

Expected evidence and useful feedback: Both recovery pathways are supported. No matched environmental winner is established. A valid grid marks missing quantities/designs, production factors, distances/modes, construction impacts, maintenance schedules, use-stage assumptions and recovery/disposal scenarios. The proposed common service is the 50 m × 6 m route for 20 years; that is a comparison brief, not a verified design.

Revise a claim when the scope changes

Expected evidence and useful feedback: A per-tonne factory result has a different unit and boundary from a whole-route life-cycle result. Different required quantities may change the comparison. A useful request specifies equal function, service period, approved design quantities, included stages, current geography-relevant data, maintenance and end-of-life assumptions, uncertainty and reviewer. Do not reward an invented emissions calculation.

Worked conclusion: "Both asphalt and concrete have documented material-recovery pathways. The supplied information does not establish which option has lower whole-life emissions for the same 50 m by 6 m route over 20 years. A comparison needs approved pavement quantities, production and transport data, construction and maintenance assumptions, and end-of-life treatment. A factory result per tonne cannot establish the proposed whole-route claim. We can request comparable evidence before recommending an environmental preference."

Evidence criteria and technical handoff

Occupational learning task	Local evidence criterion	Status after successful classroom work
A1.6 · Industry statistics	Name publisher/period/measure/scope; compute valid comparisons; distinguish employment, statistical locations and unique employers	Dated construction employment and road-industry business-location evidence demonstrated; no asphalt-only unique-employer count or current qualification outcome claimed
A1.8 · Regulations and funding	Distinguish a selected legal requirement, contract reference and public funding record; identify actual applicability questions	Document interpretation demonstrated; EPS is defined as End Product Specification in the 2006 clause; current project applicability and the historical outline's intended use of the label still need provider/context confirmation
A1.9 · Environmental comparison	Use evidenced recovery pathways, a common functional question and explicit boundaries; reject an unsupported whole-life percentage	Bounded comparison reasoning demonstrated; no project-specific environmental ranking or technical approval claimed

These are locally authored evidence criteria. The source alignment and detailed technical-review items are in the research and handoff records. Keep **Not observed** separate from **Not yet demonstrated**; provide a new dated table or a changed claim for a retry rather than asking learners to copy the key. Individual evidence includes authored text, an observed explanation, the calculation and actual digital actions where attempted.

Unit 2 · Answer and teaching guidance

Suggested 75 minutes: 10 for purpose and vocabulary, 15 for chronology, 15 for materials/categories, 15 for paired labels, 15 for independent caption and 5 for reflection. Supply H1–H4 and paper or accessible drawing tools. Model observation versus interpretation before using the photograph description. No image or private archive is required.

Build a timeline that preserves meaning

Expected evidence: 312 BCE Via Appia development begins; 1942 Alaska pioneer road completed; 1943 permanent location completed; 1946 Canadian portion responsibility transferred; 1948 civilian non-military opening; 1956 U.S. highway legislation. “Pioneer road” and “permanent location” are different milestones. Chronology alone cannot establish consultation, effects or all perspectives.

Match material, method and purpose

Expected evidence: Materials: earth, cut stone, gravel, timber logs, concrete, asphalt mixture. Asphalt cement is the bituminous binder in asphalt; Portland cement is used in the cement-based binding system in concrete. Methods: closely laid crosswise logs; delivery/placement/compaction. Purposes: moving timber, military access, travel to a community facility. Accept an explanation that different designs/materials can serve the same purpose. A photo question might concern date, location, scale, layers, use or author.

Explain a road without inventing its history

Expected evidence: Observable: logs across a route; truck in background; wet-looking appearance in the supplied description. Possible corduroy construction is a supported interpretation. Logging purpose, owner, actual moisture condition, date and location are not established. The truck does not prove logging.

Compare two explanations

Expected evidence: H2 supports the specific wartime-access explanation. It does not support a universal war-only claim; H1 and H3 also identify trade and daily movement. `1948 − 1942 = 6 years`. H2 records work/responsibility milestones within that interval.

Connect a historical method to a modern sequence

Expected evidence: One panel distinguishes timber/crosswise-log method from logging purpose. The other gives mixture delivery, paver/screed placement and roller compaction with preparation/checks acknowledged. Do not score drawing polish or accept an invented operating setting.

Independent transfer: a careful public caption

Expected evidence: A valid caption integrates a correct date with a defined event, distinguishes method/material/purpose, gives evidence for demand and names a limitation. It may discuss public-record limits without asking learners to supply cultural knowledge.

Model caption: “The Alaska Highway's pioneer road was completed in 1942 as a wartime land-access project; its permanent location was completed in 1943. Roads can later support other uses, including civilian travel and trade. Material, method and purpose describe different things: timber logs are a material, corduroy is a construction form, and logging is a road use. Modern asphalt paving combines manufactured mixture, delivery, placement and compaction. These examples show changes in techniques and needs, but they do not establish a single history for every road or describe every community's experience. A local account needs its own evidence and appropriate permission.”

Occupational learning-task evidence

Learning task	Activities	Locally authored criterion
A2.1 · Evolution of roads	“Build a timeline that preserves meaning”, “Independent transfer: a careful public caption”	Sequence dated examples and state what evolved without a universal-origin claim
A2.2 · Products and application methods	“Match material, method and purpose”, “Connect a historical method to a modern sequence”	Distinguish earth/stone/gravel/wood/asphalt/concrete and explain the binder distinction at introductory level
A2.3 · Logging and corduroy roads	“Match material, method and purpose”, “Explain a road without inventing its history”, “Connect a historical method to a modern sequence”	Distinguish purpose from construction form and avoid unsupported photographic inference
A2.4 · Modern placement	“Connect a historical method to a modern sequence”	Explain delivery, placement/screed and compaction as broad functions, with preparation/checks
A2.5 · Rationale for road growth	“Compare two explanations”, “Independent transfer: a careful public caption”	Explain a supported wartime case, post-war context and movement-of-goods purpose without a single-cause generalization

Offer a partly ordered timeline, a word bank or a dictated first draft when useful, recording the support. For a retry, change one event description or give a new category statement. A technical reviewer should check the introductory material descriptions and provider relevance; a local history contribution requires its own sourcing and permission, not instructor inference.

Unit 3 · Answer and teaching guidance

Use 15 minutes for ingredients and the flow visual, 15 for plant comparison, 15 for product clarification, 20 for recycling routes, 10 for the counter model and 15 for revision/digital note. Prepare 100 counters or a labelled paper grid; count categories, not a real mixture. Provide an actual device turn for “Make a retrievable technology note” if Digital is assessed.

Explain the ingredients to a new learner

Expected evidence: Asphalt cement and asphalt binder name the binding material in this introductory context. Aggregate is mineral material; asphalt binder binds the asphalt mixture; process fuel supplies energy and must not be assumed to be an ingredient. “Cement” may refer to asphalt cement or Portland cement. An additive may alter a specified property; its purpose and suitability require the actual product and approved mixture requirements. A pit commonly supplies unconsolidated material; a quarry commonly extracts rock that is processed. Neither source name guarantees suitable aggregate. Source suitability needs specification/material-property evidence.

Compare two process records

Expected evidence: B indicates batches; D indicates a continuous drum-mix process in the simplified case. Shared stages include supplied materials, proportioning/mixing, storage or loading and haul. A drying drum alone does not distinguish the two. Figure LA-4 shows separate weighed batches versus a continuing measured material stream; some continuous facilities use a separate dryer and continuous mixer.

Write the clarification before the order

Expected evidence: Ask for full product name/grade, current approved design or project specification, supplier/product confirmation and relevant safety/handling information through the responsible person. The undefined “Group A / C” and “altered oil” labels require an explicit clarification; do not infer a grade or product from them. No grade is approved by the task.

Trace two reuse pathways

Expected evidence: Plant route: recovery/processing → suitable plant input → approved mixture/use. Hot-in-place route: specialist coordinated process works on existing pavement in place. Both need applicable approved process, suitability and quality evidence. Removal is not proof of later reuse.

Calculate a fictional material record

Expected evidence: 80%, 15%, 5%; sum 100%. The recovered-mixture category includes more than one constituent, so the model does not isolate total binder content. The counters are not mass or a production recipe.

Respond to a changed process description

Expected evidence: Remove the unsupported plant-haul label and show the in-place route; retain the idea of recovering/reusing existing material. Identify equipment/process details as pending. Observe each learner's correction and handover.

Make a retrievable technology note

Expected evidence: The learner controls the wording, saves and retrieves a versioned file and explains the distinctions. Paper composition supports Writing but does not demonstrate save/reopen. Figure LA-3: 7 Hopper (holds supplied material); 10 Screed Extension (extends working width); 11 End Gate (helps form the mat edge); 12 High-Speed Rubber Tracks (travel component, not a speed instruction). Figure LA-5 route: hopper receives → slat conveyors move rearward → augers distribute across width → screed places/shapes the mat. Conveying rearward and distributing laterally are distinct functions. Check two selected labels against the photo. Retain model 8680, manual 1033428-SOP, printed p. 3-7 / PDF p. 43, and serial scope 365152 and above. The image does not establish operating settings, current condition or applicability to another machine.

Occupational learning-task evidence

Learning task	Activity	Locally authored criterion
A3.1 · Drum and batch plants	“Compare two process records”	Distinguish process types without classifying by one visible component
A3.2 · Ingredients	“Explain the ingredients to a new learner”, “Calculate a fictional material record”	Explain aggregate/binder/additive roles and avoid treating fuel or mixed recovered material as a single known constituent
A3.3 · Resources	“Explain the ingredients to a new learner”	Distinguish pit/quarry sources and suitability questions
A3.4 · Production liquids	“Write the clarification before the order”	Recognize that grade systems/product labels need current application and supplier confirmation; no operational selection demonstrated
A3.5 · Recycling technology	“Trace two reuse pathways”, “Respond to a changed process description”	Distinguish plant reuse, recovery and in-place processes and identify needed evidence
A3.6 · Asphalt train / hot-in-place	“Trace two reuse pathways”, “Respond to a changed process description”, “Make a retrievable technology note”	Explain a coordinated train and the distinction between placement and specialist in-place recycling

Record grade interpretation and specific plant/recycling practices as requiring qualified technical review. The classroom unit teaches the vocabulary and information decisions with complete cases; it does not establish temperatures, percentages, compatibility, settings or a machine procedure. For a retry, present a different component-based misclassification or an incomplete product request and observe the learner's reasoning.

Teach the photograph as evidence

For Figure LA-3, demonstrate one numbered callout before asking for another. A learner may point, use a magnified view, or name the callout verbally; the same source detail must be identifiable. The adjacent four-row table provides a text route for the selected component names. Keep the full manual page available when a learner asks about another number. Do not infer a function from shape alone or read the word “High-Speed” as an instruction.

The component-reading addition supports A1 equipment vocabulary and introduces I2 terminology. It is not the complete I2 technical unit. The answer-guide activity IDs and A3 criteria remain the same, so the instructor can preserve evidence continuity.

Connect the two drawings to the machine photograph

Use Figure LA-4 with “Compare two process records”, then Figure LA-5 with “Make a retrievable technology note”. Have a learner trace and say each material route before writing it. Keep the source figure/page with the explanation. The complete Understanding asphalt learner book provides MS4-03–MS4-06 for additional process comparison, saved notes and checked communication, with separate answer guidance. The source figures and original contributor credits are recorded in the adjacent captions.

Review the AI-assisted technology note and study media

The checked note must preserve the plant/process distinctions, component names, material route and unresolved product request. Inspect the actual DOCX/Markdown, PDF/HTML and SVG: a filename in a chat is not evidence of a created file. The audit must connect each technical claim to a supplied source and retain unknowns. Reject invented temperatures, mix proportions or machine settings.

The short recording must accurately distinguish hopper, slat conveyors, augers and screed; it includes a question, pause, answer and transcript. If video is created, check caption timing, readability and whether the picture supports the spoken explanation. Record the actual tool, saved formats, checks, corrections and partner’s explanation. An accurate script demonstrates script work; award recording evidence only after playback of an actual audio/video file. Observe each learner’s own file and audit actions.

Unit 4 · Answer and teaching guidance

Suggested 90 minutes: role mapping 15, safety case 20, licences 15, handover 10, schedule 15 and development/enquiry 15. Allow spoken planning before authored writing. Keep every role-play in the classroom; provide no simulated machine signals. Evaluate observable actions and source distinctions rather than confidence, accent or posture.

Make a role and dependency map

Expected evidence: Broad functions and meaningful dependencies: haul supplies mixture; paver places; screed helps control the placed layer; raker supports handwork/finishing; roller compacts; milling removes/profiles existing material. A handover identifies status or information, not an invented signal. W1 authorizes no site access or operation.

Choose the responsible next step

Expected evidence: Equipment-name knowledge is insufficient workplace-specific orientation. Learner asks for supervisor/contact, relevant hazard and emergency information, task instruction/demonstration and the appropriate communication arrangements. The employer's obligation is not transferred to the learner.

Separate four kinds of permission

Expected evidence: Admission: provider's current intake requirements. Licence: ICBC/applicable legal requirements for the actual vehicle/activity. Competence: appropriate training and assessment records. Assignment: responsible employer/site authorization. Class 5/7 in a provider prerequisite cannot authorize every truck or machine.

Repair a rushed handover

Expected evidence: Example: “The role map is ready for review, but licensing is still pending. We can print the classroom explanation with that question marked open and ask the responsible person before adding an authorization claim.” Observe the receiver's check-back and each contribution.

Check time and revise the plan

Expected evidence: 18:00–24:00 is 6 h; 00:00–02:00 is 2 h; elapsed 8 h. Subtract 0.5 h = 7.5 paid hours in the fictional record. Add Wednesday 6 h = 13.5 h. Questions may concern overtime, cancellation/travel pay, breaks, fatigue controls or actual schedule arrangements. No legal entitlement follows from these invented inputs.

Prepare a useful enquiry and growth plan

Expected evidence: Collective bargaining is negotiation between an employer and workers represented by a union about employment terms; the applicable parties, agreement and dates must be checked. Forecast demand and attendance do not guarantee hiring, continued work or qualification. Questions identify the actual agreement/terms and role-specific pathway; the goal includes a task, evidence, support, check date and responsible reviewer. A saved/reopened file is separate from a paper enquiry. Do not require sending, personal employment disclosure or joining any organization.

Occupational learning-task evidence

Learning task	Activities	Locally authored criterion
A4.1 · Health and safety	“Choose the responsible next step”	Explain the need for workplace-specific orientation and appropriate clarification
A4.2 · Paving jobs	“Make a role and dependency map”	Distinguish six broad functions and a dependency
A4.3 · Licensing	“Separate four kinds of permission”	Separate admission, licence, competence and assignment; identify verification route
A4.4 · Professionalism	“Repair a rushed handover”	Correct an overclaim, give an accurate handover and preserve pending status
A4.5 · Employer responsibility	“Choose the responsible next step”	Identify the employer's orientation/training role without assigning it solely to a worker
A4.6 · Advancement	“Prepare a useful enquiry and growth plan”	Make a role-specific learning enquiry and observable development goal
A4.7 · Job security	“Check time and revise the plan”, “Prepare a useful enquiry and growth plan”	Explain that demand forecasts and course attendance do not guarantee employment
A4.8 · Organized labour	“Prepare a useful enquiry and growth plan”	Explain collective bargaining at introductory level and check agreement parties/dates/applicability
A4.9 · Working conditions	“Check time and revise the plan”, “Prepare a useful enquiry and growth plan”	Name conditions and relevant questions without inventing employment or safety rules

Before workplace delivery, a qualified reviewer must confirm role descriptions, current legal/licensing application, provider pathway, workplace safety program and employment/collective-agreement context. This unit supplies the classroom cases and answer criteria; that review is an edition-status requirement, not a request for learners to devise actual operating procedures.

Unit 5 · Answer and teaching guidance

Use 120 minutes: project/document register 15, three-view vocabulary 25, modelled quantity 20, independent check 20, revision 25 and handover 15. Provide the complete P-01/Q-01/SP-01 text, calculator, blank paper and the cross-section visual. No physical measurements of the printed image are valid inputs. Offer unit cards and a partly completed quantity table when needed. Record support.

Sort use, ownership and authority

Expected evidence: Private facility access; Cedar Learning Society owner; Valley Surface Ltd. contractor; community fund contributor. Use/category does not by itself adopt a specification. Actual signed contract and authorized current documents determine applicability.

Choose the right view

Expected evidence: Plan: 50 m horizontal length/6 m width. Profile: elevation change along centreline. Section: 40 mm calculation-layer thickness. 100.50 m is an elevation relative to the illustrative datum, not a layer depth.

Describe the road form

Expected evidence: Crown raised transverse form; shoulder road edge area; open ditch drainage channel; curb/gutter contained edge/drainage arrangement; sidewalk pedestrian route; runoff surface-water movement; superelevation cross-slope arrangement on curves; grader equipment. Figure LA-6: granular base supports the asphalt surface; subbase is an additional supporting layer where specified; prepared subgrade is the prepared ground beneath the pavement structure. Other structures exist; the drawing supplies no P-01 design dimensions. Questions may concern outlet, base preparation, approved slope, services or required materials. Do not supply invented design answers.

Check the quantity and record

Expected evidence: Thickness 0.040 m; area 300 m²; volume 12 m³; model mass 24 t. Records 8 + 8 + 7.6 = 23.6 t, 0.4 t below the model quantity. Profile rise 0.50 m/50 m=1%. Neither record totals nor an invented factor establish actual placed quantity, thickness, density, acceptance or payment.

Recalculate after the revision

Expected evidence: Revision C area 360 m²; volume 14.4 m³; model mass 28.8 t. Increases 60 m², 2.4 m³, 4.8 t. Existing record total 23.6 t is 5.2 t below the revised model quantity. Profile rise 0.60 m/60 m=1%. Keep the old record and identify which revision each calculation uses.

Review and hand over

Expected evidence: Handover includes current revision C, 14.4 m³ geometric model, 28.8 t at fictional 2.00 t/m³, 23.6 t recorded, 5.2 t arithmetic difference, and the limits on a real order or acceptance decision. Each person demonstrates both checks over the two cases.

Model quantity note: “P-01 revision C shows a 60 m by 6 m rectangle with a 0.040 m calculation layer. Area is 360 m² and geometric volume 14.4 m³. At the supplied fictional 2.00 t/m³ factor the model mass is 28.8 t. The unchanged classroom records total 23.6 t, a 5.2 t difference. I checked units and recalculated from the new length. These model values do not establish an actual order, approved density or acceptance result.”

Occupational learning-task evidence

Learning task	Activities	Locally authored criterion
A5.1 · Project categories	“Sort use, ownership and authority”	Distinguish use from jurisdiction and avoid treating categories as mutually exclusive
A5.2 · Owner types	“Sort use, ownership and authority”	Recognize provincial/federal/municipal/private possibilities and distinguish owner/funder/contractor in the case
A5.3 · Common work/forms	“Choose the right view”, “Describe the road form”	Describe listed road forms, drainage vocabulary and grader-equipment distinction
A5.4 · Specifications	“Sort use, ownership and authority”, “Recalculate after the revision”	Find the applicable exercise document and revision; identify real contract applicability as a separate check
A5.5 · Cross-section/preparation	“Choose the right view”–“Check the quantity and record”	Read conceptual layers and identify unanswered civil/preparation questions without treating the sketch as design
Bridge toward F2/F3	“Check the quantity and record”–“Review and hand over”	Demonstrate drawing-reading and quantity arithmetic with stated assumptions; no F2/F3 practical competence or road-design approval claimed

The technical reviewer must confirm the conceptual road terminology, drawing conventions, selected quantity model, appropriate metric/imperial bridge and any later use of real specifications or acceptance limits. Do not substitute these classroom numbers for a project design. A retry can change one dimension, record unit or revision; keep enough information to calculate and explain the result independently.

Unit 6 · Answer and teaching guidance

Use 90 minutes: claim reading 15, material/lightweight-fill discussion 15, cost calculation 10, supplier enquiry 10, option table 20, changed condition 10 and brief/check 10. Supply N1–N8 and the retained public-source excerpts. Learners need no product samples, site data or external accounts. Distinguish an evidence-supported mechanism from a project performance prediction.

Improve three claims

Expected evidence: Paving-stone permeability depends on the complete system; porous asphalt with a reservoir still needs site/drainage design and maintenance; recycling alone cannot establish total impact. Each revision names a condition and a question rather than simply adding “may” to a vague claim.

Read the cost boundary

Expected evidence: `$1,145,000 − $633,000 = $512,000`. Limits include historical 2016/2017 context, California/U.S. geography/currency context, project-specific quantities/design, missing installation/overhead/profit and no current B.C. supplier quotes. Technical suitability depends on site/design requirements, not the cheapest listed total. The example concerns lightweight fill; that role cannot be relabelled as approved pavement sub-base without design and material evidence.

Make a product enquiry

Expected evidence: Questions identify actual product/system, intended use, relevant tested performance, maintenance, compatibility and approval. Pattern questions include access/visibility and permission for any protected imagery. No universal epoxy, colour or pattern performance is accepted without evidence.

Compare options against the actual need

Expected evidence: Porous asphalt relates to water management, subject to site/loading/drainage/maintenance study. Coloured asphalt may relate to wayfinding; it does not answer infiltration or structural needs by colour alone. EPS geofoam relates to embankment/load conditions that have not been established for this courtyard. Accept different investigation shortlists when evidence and limits are explicit. Appropriate review may include qualified pavement/civil/geotechnical/design professionals and the facility's maintenance decision-maker.

Adapt to a new constraint

Expected evidence: Lack of a specialized-maintenance commitment is relevant to clogging/maintenance risk in the porous-asphalt reference. A reasonable revision makes maintenance feasibility a condition for investigation or postpones that option. It does not establish an alternative's suitability.

Build an evidence brief and test its clarity

Expected evidence: The brief recommends an investigation, with a known mechanism, a case-specific unknown, source date and next decision. Reader can distinguish a recommendation to study from permission to construct. Record actual file actions separately.

Model brief: “For the fictional 600 m² courtyard, investigate whether a porous-asphalt/stone-reservoir system could meet the surface-water and loading requirements. FHWA's2015 brief describes the mechanism and identifies clogging and specialized maintenance as considerations. We have no infiltration/geotechnical results or maintenance commitment, so we cannot recommend construction. First obtain site/design review and a feasible maintenance plan. Evaluate wayfinding separately: colour may help a communication purpose but does not establish drainage performance. Keep environmental comparisons within a defined life-cycle boundary and use project-specific information before ranking options.”

Occupational learning-task evidence

Learning task	Activities	Locally authored criterion
A6.1 · Green roads: paving stones, porous asphalt, recycling	“Improve three claims”, “Compare options against the actual need”	Explain three different mechanisms and conditions; avoid universal “green” claims
A6.2 · Epoxy/coloured/modified materials	“Make a product enquiry”, “Compare options against the actual need”	Distinguish product families and request performance/application evidence
A6.3 · New sub-bases/pumice	“Read the cost boundary”	Identify pumice in a bounded historical lightweight-fill comparison and state suitability/data limits
A6.4 · Floating roads/foam	“Read the cost boundary”, “Compare options against the actual need”	Explain lightweight-fill purpose and EPS term without implying flotation or generic-foam substitution
A6.5 · Added-value enhancements/street print	“Make a product enquiry”, “Build an evidence brief and test its clarity”	Separate visual purpose, technical evidence and permission

A qualified reviewer must check the material descriptions, local suitability questions, dated-reference use and terminology before technical delivery. These sources are evidence for mechanisms and questions, not current B.C. design approval. A retry can use an unfamiliar marketing claim with a different missing condition. Ask learners to identify what evidence would change their conclusion.

Additional worked answers and assessment guidance

A1 · Foundations added to Chapter 1

Build the material story

Expected evidence and reasoning: Source: approved aggregate/binder supply and identity; production: plant and specified mixture record; delivery: haul vehicle/ticket; placement: paver/tractor/screed and current work location; compaction/checking: roller and applicable inspection/test evidence. Accept equivalent accurate terms. Look for each learner explaining the handover, not simply copying five boxes.

Connect a part to a purpose

Expected evidence and reasoning: Correctly locate the two paver sections and eight requested hand-tool features. Compare the real product rather than assuming every lute/rake has the same head. A usable report: “During the teaching inspection, the rake head moved at the handle connection. I set it aside from use and reported it to the instructor for assessment under the tool procedure.” A repair specification is not available.

Classify carefully

Expected evidence and reasoning: Gravel/concrete: surface or material; logging/recreational access: use; corduroy: construction form; seasonal ice road: surface plus seasonal context. Roads can support services, trade, work and recreation. Appropriate project questions concern access changes, land/water effects and who must be consulted. No learner is required to disclose cultural knowledge.

Give a complete first explanation

Expected evidence and reasoning: A coherent route uses the five required terms accurately, explains a purpose and supporting layers, and identifies one legitimate limit such as actual thickness, density or operating settings. Observe the saved/reopened file for Digital. Record the listener's restatement and the author's revision.

A3 · Machine and process studies

Trace movement and identify parts

Expected evidence and reasoning: Feeder bed plate/chain/foot shaft/head shaft in the feeder diagram; auger segments/shaft in the auger diagram. Feeding moves mixture rearward; augers distribute it laterally. The bed plate is a wear surface beneath the material path; the chain/shaft assembly has moving parts.

Actual service work requires the relevant isolation and maintenance procedure.

Explain a relationship without guessing a setting

Expected evidence and reasoning: Match forward motion, screed weight, head of material, angle of attack and material stability. Weight is one factor in a responding system; it does not guarantee a fixed layer thickness. Identify actual model/manual, task conditions, approved procedure and qualified instruction before changing controls. No numeric setting can be inferred.

Check the claim in a roller record

Expected evidence and reasoning: P has two recorded passes. Right strip has none, so complete width coverage cannot be claimed. The record does not establish density or an accepted compaction result. A clearer record distinguishes point and width observations and links the required test evidence. No operating pattern should be prescribed.

A5 · Map desk

Read, toggle and preserve a record

Expected evidence and reasoning: Site point: location record, not entrance or boundary. Entrance: teaching access point, not authorization for a real vehicle or route. Utility record: approximate and unverified, not exact position, depth or excavation clearance. Hidden layer changes display only. Observe layer toggling, record selection, export and reopening; check revision and selected record in the exported JSON.

Check the revision in two places

Expected evidence and reasoning: B: 50 × 6 = 300 m²; 40 mm = 0.040 m; volume 12 m³; model mass 24.00 t. C: 60 × 6 = 360 m²; volume 14.4 m³; model mass 28.80 t. Increases: 60 m², 2.4 m³, 4.80 t. Delivered record total remains 23.60 t. Difference below model requirement: 0.40 t for B, 5.20 t for C. The factor is fictional and the difference does not establish placed thickness, density or payment.

Make a usable access brief

Expected evidence and reasoning: Distinguish all four features, name the selected revision and identify genuine information gaps. Accept different clear prose. Reject claims that a parcel gives permission, a map line gives a safe dig location or a car route permits a heavy truck. Confirm an individual authored brief and revision.

Worked access brief (revision C): “Cedar Learning Access uses P-01 revision C for this classroom calculation. The site point locates the fictional facility; the entrance point identifies its teaching access location. The dashed boundary and approximate utility record have different meanings. Neither gives permission to excavate or establishes a verified utility alignment. The model route is 60 m by 6 m with a

0.040 m layer: 14.4 m³ and 28.80 t using the supplied invented factor. Actual work instructions would need the authorized design, verified access and utility arrangements, applicable traffic plan and current site controls. Please confirm the responsible reviewers before any field instruction is issued.”

B1 · Safety chapter

Follow the responsibility without losing the hazard

Expected evidence and reasoning: 8.2 lists weather clothing, general-purpose gloves, appropriate/safety footwear and safety headgear as worker provision responsibilities; agreements may allocate supply to the employer. Employer supplies other required PPE without charge. Purchasing an item does not prove selection, fit, compatibility, serviceability or training.

Inspect the decision, not just the item

Expected evidence and reasoning: All four cases have unresolved protection. Avoid the exposure while the responsible person resolves selection/fit or withdraws defective/contaminated items under the procedure. Respirator interference needs the respiratory program's response, not an improvised adjustment. Chemical glove suitability requires product/task evidence; cracked headgear and contaminated equipment require appropriate removal/assessment/care.

Build a hazard-and-control record

Expected evidence and reasoning: (a) Possible inhalation/combustion exposure; assess equipment, location, ventilation and contaminants; do not treat a confined space as ordinary open work. (b) Ingestion route; hygiene and separation of contaminated work from eating. (c) Force/repetition/posture plus trip/route hazards; assess tool/load/layout/work organization. (d) Heat/workload/acclimatization concerns; applicable plan, controls and supervision. Identify affected people and a responsible role, with reassessment after a change. No exposure limit or control effectiveness can be inferred from the brief.

Reassess when the conditions change

Expected evidence and reasoning: Wind and access changes affect exposure and affected people. Withdraw the old conclusion; have the relevant people reassess task controls and the entrance arrangement. An improvement makes unresolved decisions and responsibilities easier to find. Record each learner's contribution and the result of the peer test.

Close the orientation gaps

Expected evidence and reasoning: O-01 authorizes only the stated supervised teaching task; actual machine operation is excluded. Real emergency/first-aid/assembly details and demonstration limits require confirmation. Questions should be specific and assigned to the instructor/venue responsible person. Invented emergency details are unacceptable.

Resolve a label-and-SDS mismatch

Expected evidence and reasoning: Product identifiers do not match; prevent use and obtain the correct label/SDS and workplace instruction through the responsible person. A similar name is insufficient. Observe actual file retrieval/save/reopen. Pictogram meanings must match CCOHS; no pictogram alone selects PPE or provides a complete emergency response. Environmental symbol is not mandatory under WHMIS.

Make the emergency message useful

Expected evidence and reasoning: The learner records verified venue arrangements or the responsible person and open confirmation. Assess the spoken message for location, incident, people and known hazards, without a real call. Classroom practice is not first-aid certification; actual designated coverage follows the employer assessment and applicable requirements.

Read a proposed lift for missing evidence

Expected evidence and reasoning: Missing load mass, lift points, legible equipment identity/WLL, suitable configuration, inspection, qualified rigger/supervision and communication arrangement. No lift should proceed on this information. 15.2 concerns qualified people; 15.31 requires visual inspection before use each shift and immediate removal of defective gear; 15.34 addresses sling angle from vertical and required WLL reduction.

Identify geometry and communicate clearly

Expected evidence and reasoning: Identify the angle between sling leg and vertical, not horizontal. Signal names must match the official picture; verify each chosen signal directly. Unclear/conflicting signals require the response in the actual safe-work procedure and qualified communication arrangements. The exercise demonstrates recognition, not operational signaller competence.

Use the manual to ask the right question

Expected evidence and reasoning: Use the actual warning text in the illustrated CB2.5 GC/CB2.7 GC/CC2.7 GC manual, with publication and serial-range check. Identify the steering frame lock and articulation area. A stopped engine or one lock does not prove electrical, hydraulic, thermal, gravity and other energy is controlled. Actual isolation/blocking/verification is procedure-specific.

Stop an unsupported site assumption

Expected evidence and reasoning: Map: obtain actual utility responses/locates/procedure. Overhead: confirm line/voltage and applicable limits/procedure with the responsible authority. Raised part: avoid exposure and have trained people establish specified support/isolation. Familiar task: reassess current conditions and instruction needs. Digital evidence is the saved plan-record note.

Review the public route

Expected evidence and reasoning: Consider facility users, deliveries, mobility/access needs and nearby workers. Drawing lacks actual dimensions, traffic-control plan, barrier specification and site assessment. Questions address safe accessible route, visibility, entry control, emergency access and change management. A notice can be clearer without pretending to design the work zone.

Integrated classroom safety brief

Expected evidence and reasoning: All nine requested brief elements appear, are internally consistent and identify responsibility/evidence for open decisions. A strong brief does not mark unverified work “safe.” Peer review tests findability; final record includes individual reasoning, revisions and observed digital actions.

Assess the integrated brief fairly

Use **Not observed**, **Not yet demonstrated**, **Demonstrated with support** and **Demonstrated independently** for the task under the observed conditions. Record the support and next practice. These local descriptions are not official Skills for Success levels or a trade qualification.

Skill	Individual evidence to observe in “Integrated classroom safety brief”
Reading	Locates a clause, manual warning or SDS requirement and preserves its condition.
Writing	Produces a clear brief with named responsibilities and unresolved items.
Numeracy	Retains the checked P-01 quantity/revision and distinguishes angle conventions; does not invent a lifting capacity.
Digital	Saves, exports or organizes the brief and reopens the correct version.
Problem Solving	Identifies a missing fact and the appropriate route to resolve it.
Communication	Explains a decision and confirms the listener's understanding.
Collaboration	Gives and uses specific peer feedback while retaining individual evidence.
Adaptability	Revises an assessment after a new condition or corrected record.
Creativity and Innovation	Tests a useful visual or record improvement against a reader's needs.

Numeracy inclusion for the integrated brief: attach the checked Chapter 5 quantity/revision sheet and the marked angle-convention diagram from “Identify geometry and communicate clearly”. This keeps a visible individual numeracy product with the brief. It does not add a lift calculation.

For delivery, divide Chapter 7 into supported sessions with time for reading, discussion and individual attempts. Use pictures or isolated teaching samples for recognition exercises. Schedule qualified instruction and authorized practical observation separately. Before teaching, verify current legislation,

actual venue emergency arrangements, the fleet/manual match and any product-specific SDS. Technical approval and classroom pilot remain open in the delivery record.

Coverage and delivery record

All 61 later B2–L6 competencies now have complete classroom chapters in Volumes 2–8; see the 218-task coverage map. Technical review and observed practical assessment remain separate.

Competency	Learning tasks covered	Main evidence
A1 · Industry and equipment	All nine: process/equipment, history, importance, aggregate resources, uses, statistics, road types, regulations/funding and environmental comparison	“Build the material story” through “Give a complete first explanation”, “Report what the data can support” through “Revise a claim when the scope changes”; Chapters 2–3 develop history/materials
A2 · History of asphalt pavement	All five: evolution, products, logging roads, placement methods and road demand	“Build a timeline that preserves meaning” through “Independent transfer: a careful public caption”
A3 · Technology of asphalt pavement	All six outline learning tasks; process and material distinctions, plant routes, paving/compaction and current methods	“Explain the ingredients to a new learner” through “Make a retrievable technology note”; “Trace movement and identify parts” through “Check the claim in a roller record” add component/relationship studies
A4 · Workers in the paving industry	All nine outline learning tasks, including roles, responsibilities, licensing, professionalism, development and organized labour	“Make a role and dependency map” through “Prepare a useful enquiry and growth plan”
A5 · Road and highway profiles	All five: project categories, ownership, work types, specifications and cross-sections	“Sort use, ownership and authority” through “Review and hand over”; “Read, toggle and preserve a record” through “Make a usable access brief” extend map records and digital quantities
A6 · Emerging trends	All five outline learning tasks, with product/process terms, evidence boundaries and comparison practice	“Improve three claims” through “Build an evidence brief and test its clarity”
B1 · Safety orientation	All seven: PPE; hazard assessment; company orientation; WHMIS/first aid; rigging; site risks; public safety/security	“Follow the responsibility without losing the hazard” through “Integrated classroom safety brief”

The chapter tables connect learning tasks to practice; the complete outline wording and page locations remain available through the full audit. This table describes teaching coverage, not occupational competence awarded.

Before delivery

Check	Current record
Source identity	Supplied outline/profile verified against current official downloads in the source audit.

Check	Current record
MS-4 integration	Actual source crops, verified redraws, component explanations and figure/page credits included.
Machine applicability	Manufacturer guide and OMM identities retained; actual training fleet models/serials must still be matched.
Legislative review	Current retained WorkSafeBC articles consulted on 30 October 2026; confirm any later changes before delivery.
Venue readiness	Instructor must confirm emergency arrangements, practical boundaries, equipment and supervision.
Technical approval	Pending qualified provider/trade review.
Classroom pilot	Pending; collect clarity, access, timing and assessment evidence from actual learners.
Indigenous/local review	Invite the appropriate relationships and local review for the delivery setting; no Nation-specific teachings invented.

Continuity with earlier resources

The six existing roadbuilding chapters and their worked answers have been brought into this study volume and expanded. Activity identifiers remain in authoring records and HTML metadata; reading editions use descriptive task names. The six inherited LA figures keep their existing credits; the additional figure register supplies machine-readable provenance and source/crop hashes.

CONSULTING contributes authored map-reading and source-quality workflows, pinned to a verified commit. The lesson adapts these to fictional practice geometry. The original documentation snapshots are retained for audit, while the learner chapter explains the concepts directly.